6.1 Flow Design

To effectively address the opera tional challenges identified in Milestone 2 (**B2: Unstructured Dispatch** and **B3: Post-Resolution Data Latency**) while supporting the business goals established in Milestone 1, GreenGrid Utilities adopts a configuration-driven automation strategy using **Salesforce Record-Triggered Flows**.

By eliminating manual coordination processes and paper-based reporting, the platform enables smooth, real-time integration between customer service operations and field service activities.

6.1.1 Flow 1: Automated Field Dispatch Trigger (Dispatch Engine)

- **Triggering Object:** Case
- **Trigger Source Event:** Record creation or update
- **Trigger Condition Criteria:** Priority EQUALS High AND Status EQUALS New AND Asset_Lookup__c IS NOT NULL TRUE
- **Optimization Goal:** Rapid Field-Service Triage (Actions & Related Records)

Architectural Overview:

When a customer service representative logs an outage request and creates a High-Priority Case associated with a valid Meter ID, this flow automatically removes the need for manual dispatching. It creates a related Field_Visit__c record, connects it to the parent Case, transfers the required customer reference data, and immediately sends a system notification to the field service team.

6.1.2 Flow 2: Real-Time Operational Sync and Case Closure (Sync Engine)

- **Triggering Object:** Field Visit (Field_Visit__c)
- **Trigger Source Event:** Record update
- **Trigger Condition Criteria:** Repair_Status__c EQUALS Resolved AND Repair_Status__c ISCHANGED TRUE
- **Optimization Goal:** Instant Operational Dashboard Visibility (Actions & Related Records)

Architectural Overview:

Once a field technician submits repair details through the Salesforce Mobile App and changes the status to "Resolved," this automated process transfers the technician's resolution notes to the related Case record. The flow then updates the Case status to "Closed" and records the completion timestamp, removing reporting delays and providing immediate visibility into operational performance.

6.2 Workflow Documentation

6.2.1 Core Functional Workflows

```
[Customer Support Agent Saves High-Priority Case]
                      |
                      ▼
        (Flow 1: Dispatch Engine Executes)
                      |
                      ▼
          [Creates Field_Visit__c Record]
                      |
                      ▼
 [Sends Push Notification to Field Tech Mobile App]
                      |
                      ▼
          [Field Tech Resolves On-Site Issue]
```

|
▼
(Flow 2: Sync Engine Executes)

|
▼
[Updates Parent Case Status to "Closed" & Logs Notes]
|
▼
[Executive Dashboard Updates in Real-Time]

a. Outage Intake to Automated Technician Dispatch (Flow 1 Workflow)

1. **User Action:** A Customer Support Agent receives an outage report, performs a Meter ID lookup to associate the Utility_Asset__c record, completes the Case form with Priority = High and Status = New, and saves the record.

2. **Automated Evaluation:** The Salesforce processing engine validates the transaction and confirms that the Case meets the criteria defined for Flow 1.

3. **Record Generation:** The flow creates a new Field_Visit__c record, populates Parent_Case__c with the Case ID, assigns Repair_Status__c = Assigned, and references the related customer information.

4. **Notification Routing:** The platform sends an immediate desktop and mobile notification to the designated field service queue.

b. Mobile On-Site Resolution to Automated Case Closure (Flow 2 Workflow)

1. **User Action:** After completing the repair work, the Field Technician accesses the record through the Salesforce Mobile App, enters the required Technician_Notes__c information, updates Repair_Status__c to “Resolved,” and saves the record.

2. **Automated Evaluation:** Salesforce immediately launches Flow 2 when the status change is detected.

3. **Upward Record Update:** The flow updates the related parent Case through Parent_Case__c, sets Status = Closed, and copies Technician_Notes__c into the Case resolution information section.
4. **Metrics Capture:** The flow records Resolution_Timestamp__c on the Field Visit record to support executive reporting and dashboard analytics.

6.3 Error Handling Strategy

To maintain 99.9% platform availability (**NFR1**) and ensure that no critical outage records are lost because of unexpected transaction failures, a structured error-management mechanism is incorporated into every flow design.

6.3.1 Fault Path Routing

Each DML (Data Manipulation Language) operation—including **Create Records** and **Update Records** elements—is configured with a dedicated **Fault Path Connector**. If issues such as record locking, validation rule failures, or database timeouts occur, Salesforce captures the exception and handles it gracefully without displaying a generic application error to the user.

6.3.2 Exception Notification and Automated Logging

Whenever a fault path is triggered, the platform performs two automated error-management procedures:

1. **System Administrator Alerting**

   A standard **Send Email** action automatically notifies the IT System Administrator group. The email includes important troubleshooting information such as the Record ID, initiating user profile, execution timestamp, and the system-generated error message ($Flow.FaultMessage).

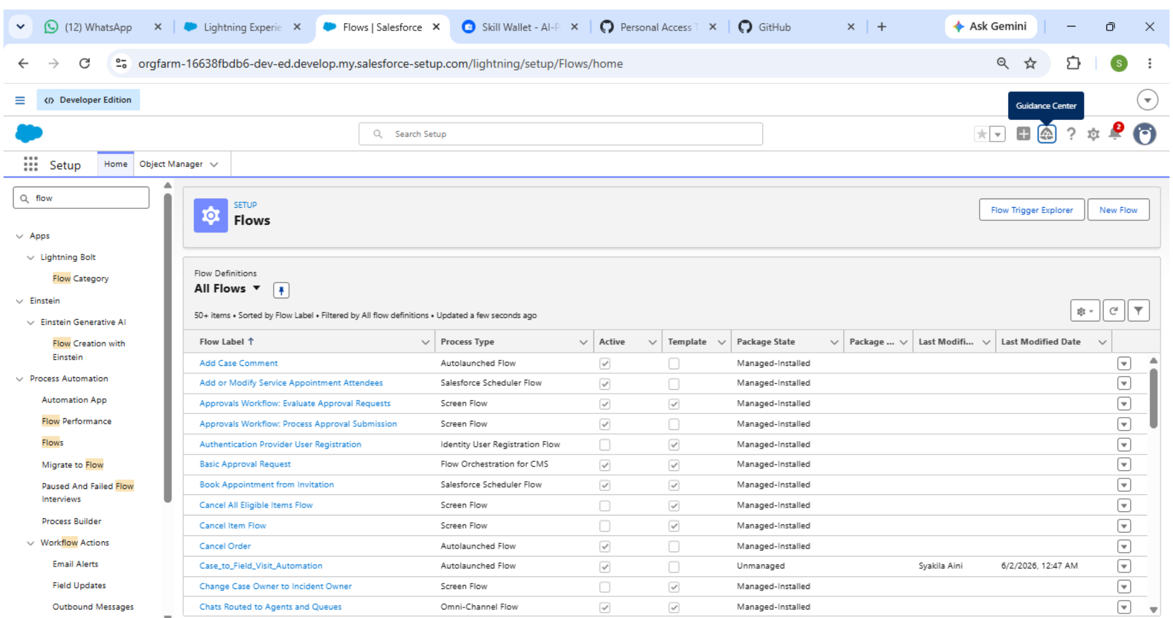
2. **Chatter Fallback Notification**

   The flow also publishes a high-priority message to a restricted IT System Administrator Chatter group, ensuring issue visibility and traceability even when email delivery is delayed because of network interruptions.

6.4 Salesforce Implementation and Configuration Steps

Step 1: Open the Flow Builder Workspace

1. Sign in to your Salesforce Developer environment.
2. Select the gear icon in the upper-right corner and choose **Setup**.
3. Enter **Flows** in the **Quick Find** search box.
4. Select **Flows** from the *Process Automation* section.



Step 2: Create Flow 1 — Automated Field Dispatch Trigger

1. On the Flows landing page, click **New Flow**.

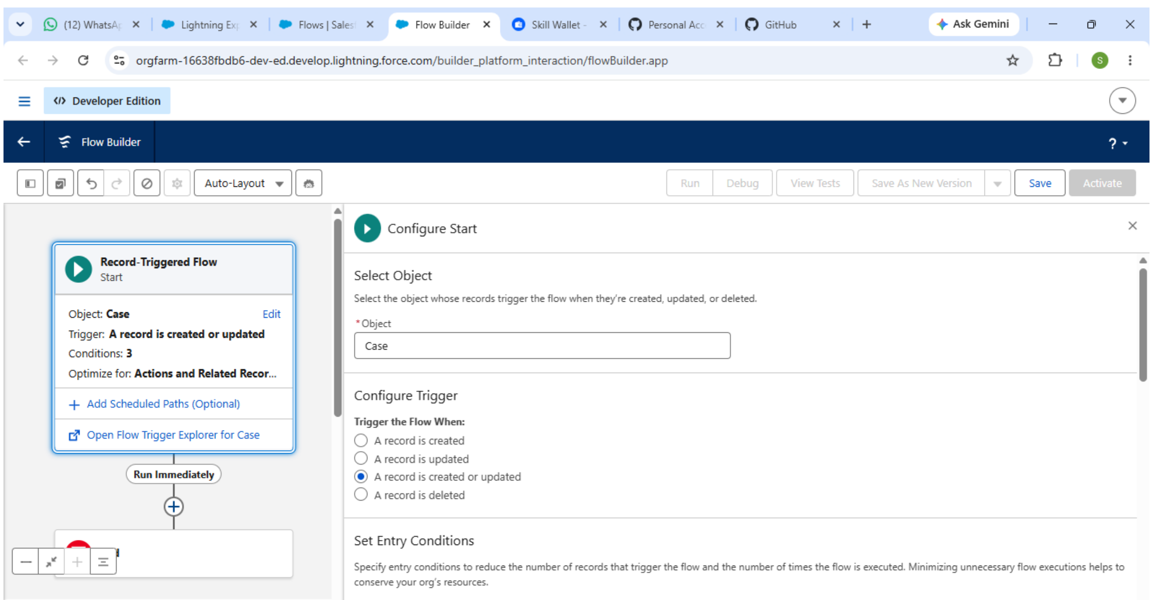
2. Choose **Record-Triggered Flow** and click **Create**.

3. Configure the **Start** settings as follows:

- **Object:** Case
- **Trigger the Flow When:** A record is created or updated
- **Optimize the Flow for:** Actions and Related Records

4. In the **Set Conditions** section, configure:

   - **Condition Requirements:** All Conditions Are Met (AND)
   - **Field 1:** Priority | **Operator:** Equals | **Value:** High
   - **Field 2:** Status | **Operator:** Equals | **Value:** New
   - **Field 3:** Utility_Asset__c | **Operator:** Is Null | **Value:** $GlobalConstant.False



5. Under **When to Run the Flow for Updated Records**, select:

   - Only when a record is updated to meet the condition requirements.

Close the configuration window.

Step 3: Add a Create Records Element for Field Visit Creation

1. Click the plus (+) icon beneath the Start element.
2. Search for and select **Create Records**.
3. Configure the element:

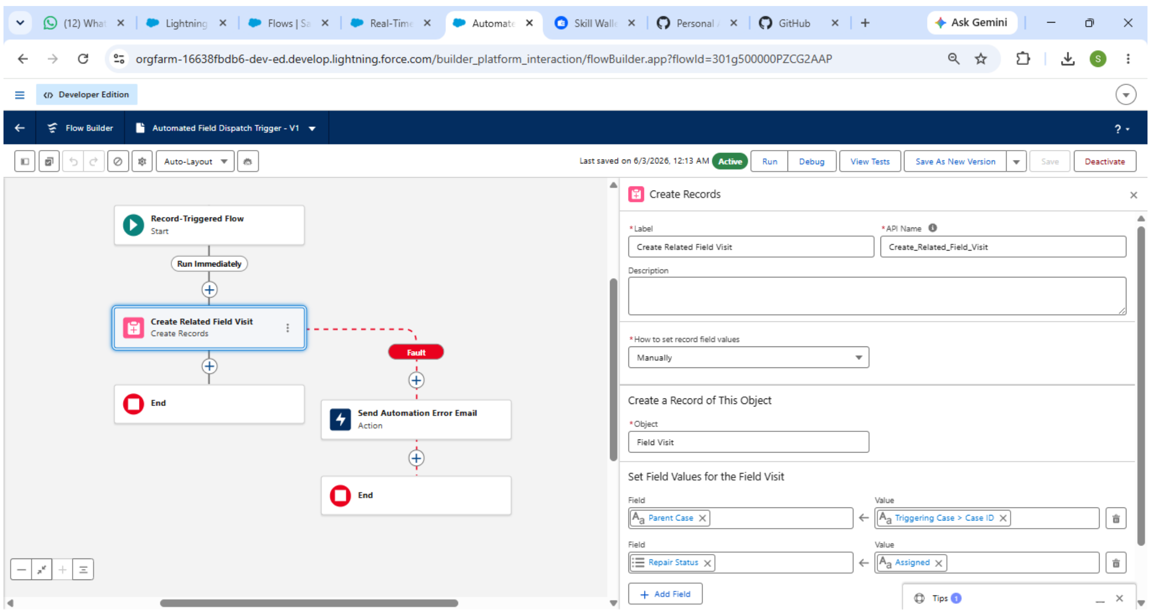
- **Label:** Create Related Field Visit
- **API Name:** Create_Related_Field_Visit
- **How Many Records to Create:** One
- **How to Set the Record Fields:** Use separate resources and literal values

4. Choose:

   - **Object:** Field_Visit__c

5. Configure field mappings:

   - **Parent_Case__c** ← {!$Record.Id}
   - **Repair_Status__c** ← Assigned



6. Save the configuration.

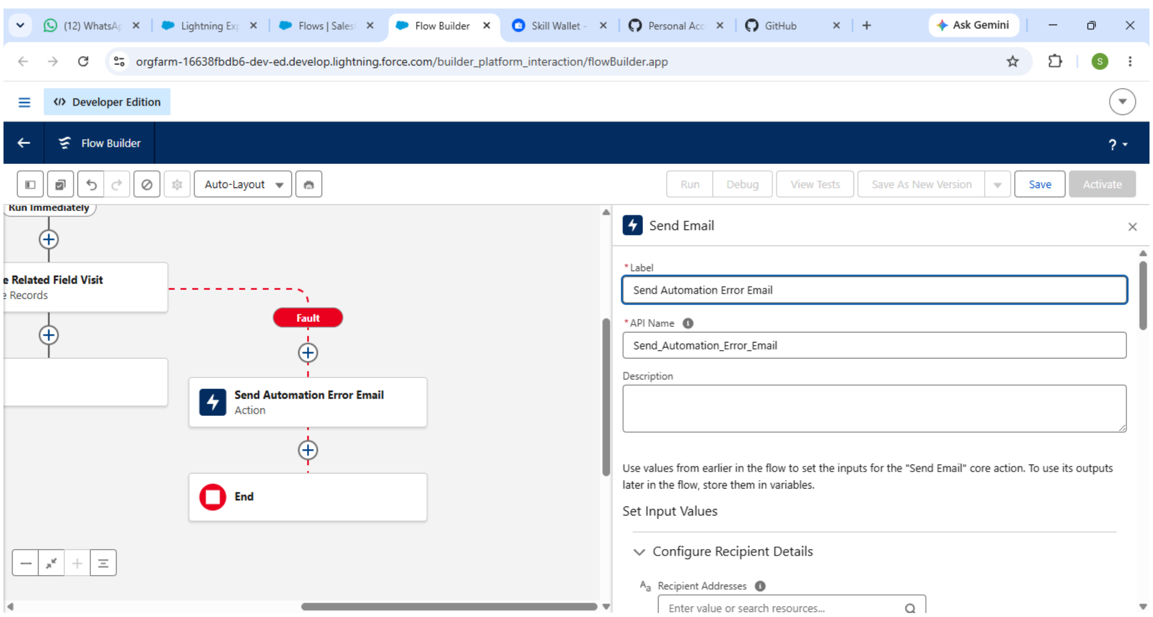
Step 4: Configure Fault Path Error Handling for Flow 1

1. Select the **Create Related Field Visit** element.
2. Choose **Add Fault Path**.
3. Click the plus (+) icon on the Fault Path branch.
4. Select **Action**.

5. Search for **Send Email** and select the standard email action.

6. Configure:

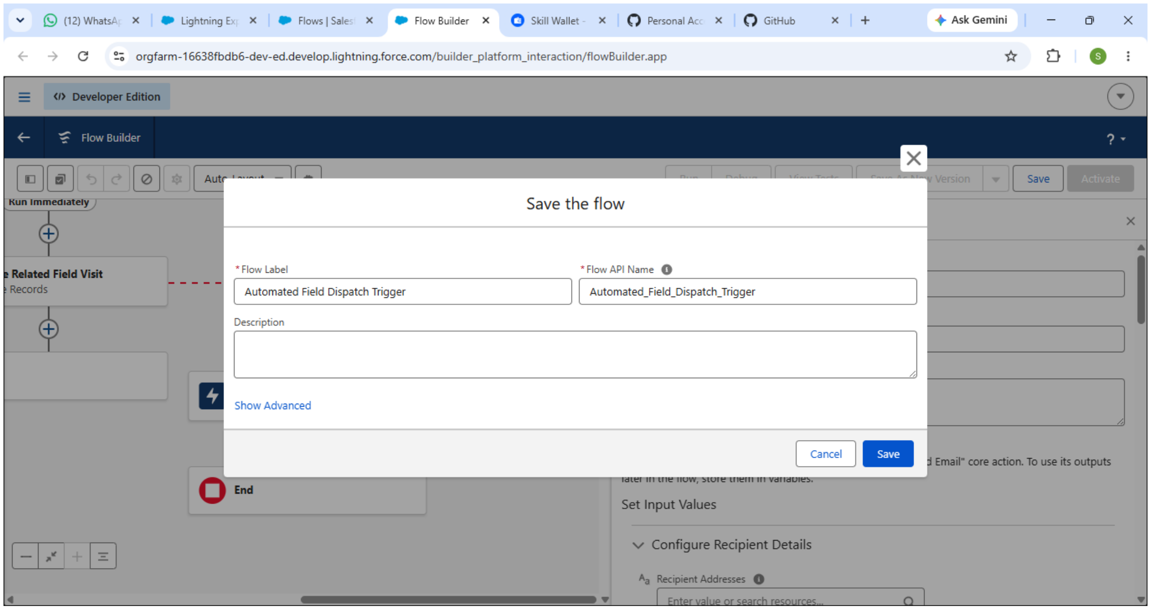
   - **Label:** Send Automation Error Email
   - **API Name:** Send_Automation_Error_Email
   - **Body:** The Field Dispatch Flow failed to create a technician dispatch record. Error Message details: {!$Flow.FaultMessage}
   - **Subject:** CRITICAL ALERTER: GreenGrid Utilities Salesforce Automation Failure
   - **Email Addresses:** Enter the administrator email address.
   - **Target Workflow Sender:** Set to True



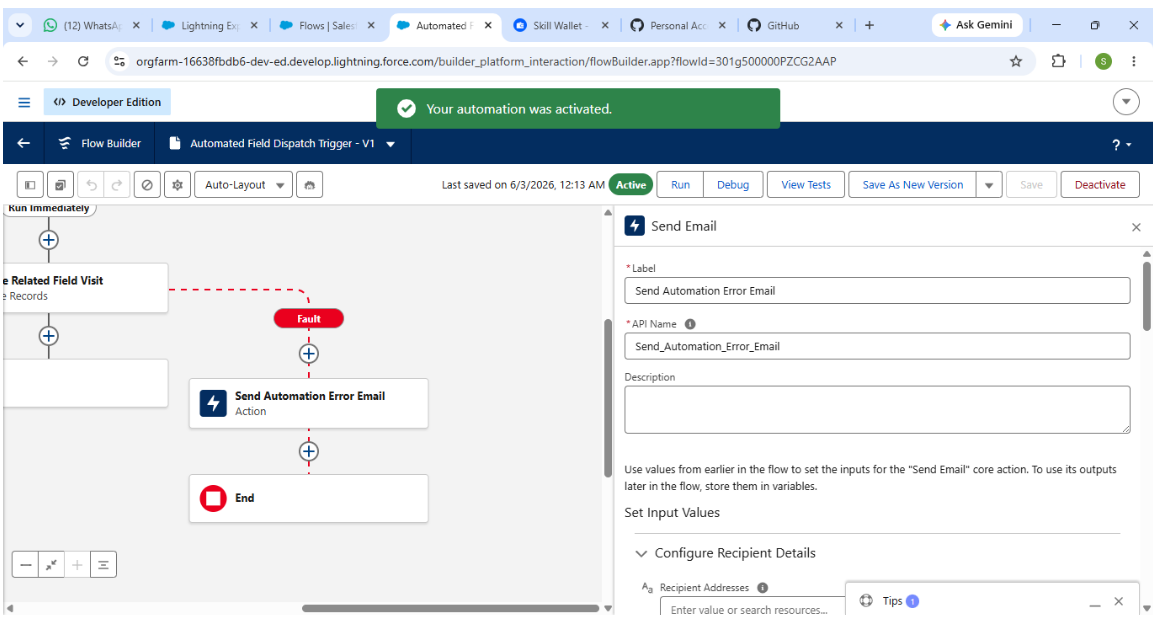
7. Click **Done**.

Step 5: Save and Activate Flow 1

1. Click **Save**.
2. Set the **Flow Label** to *Case: Automated Field Dispatch Trigger*.
3. Confirm the **Flow API Name** is *Case_Automated_Field_Dispatch_Trigger*.



4. Click **Save**.



5. Select **Activate**.

6. Return to Setup.

Step 6: Create Flow 2 — Real-Time Operational Sync and Case Closure

1. From the Flows page, click **New Flow**.

2. Select **Record-Triggered Flow** and click **Create**.

3. Configure:

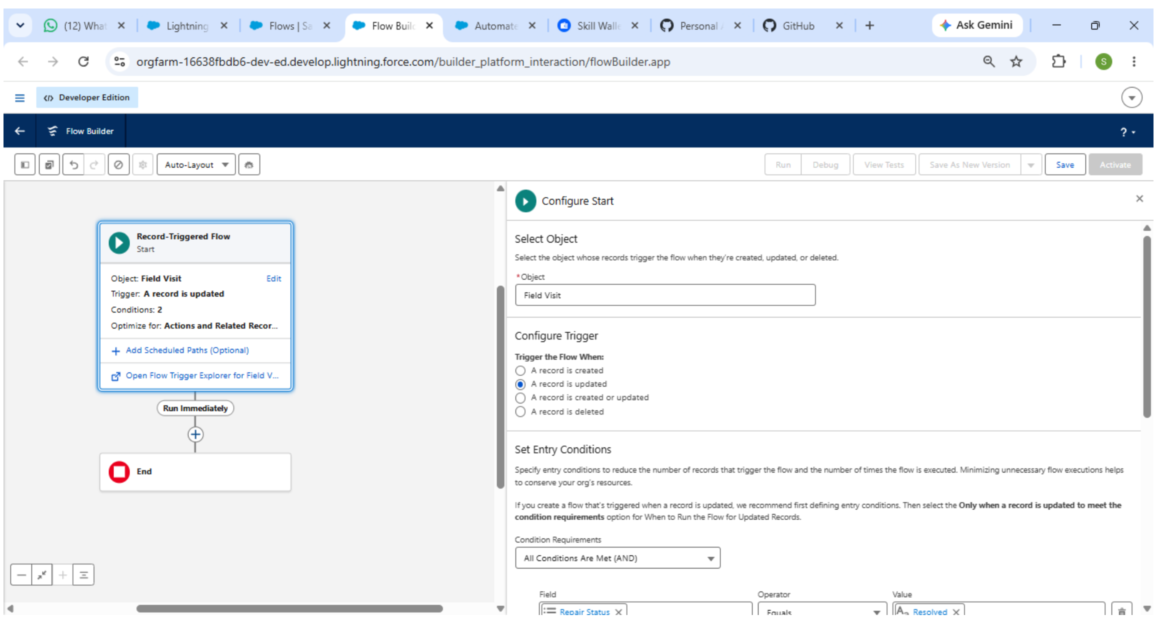
   - **Object:** Field_Visit__c
   - **Trigger the Flow When:** A record is updated
   - **Optimize the Flow for:** Actions and Related Records

4. Set the following conditions:

   - **Condition Requirements:** All Conditions Are Met (AND)
   - **Field 1:** Repair_Status__c | **Operator:** Equals | **Value:** Resolved
   - **Field 2:** Parent_Case__c | **Operator:** Is Null | **Value:** $GlobalConstant.False

5. Choose:

   - Only when a record is updated to meet the condition requirements.



Close the configuration window.

Step 7: Add an Update Records Element for Case Synchronization

1. Click the plus (+) icon below the Start element.
2. Select **Update Records**.
3. Configure:

- **Label:** Update Parent Case to Closed
- **API Name:** Update_Parent_Case_to_Closed
- **How to Find Records to Update and Set Their Values:** Specify conditions and set fields individually

4. Choose:

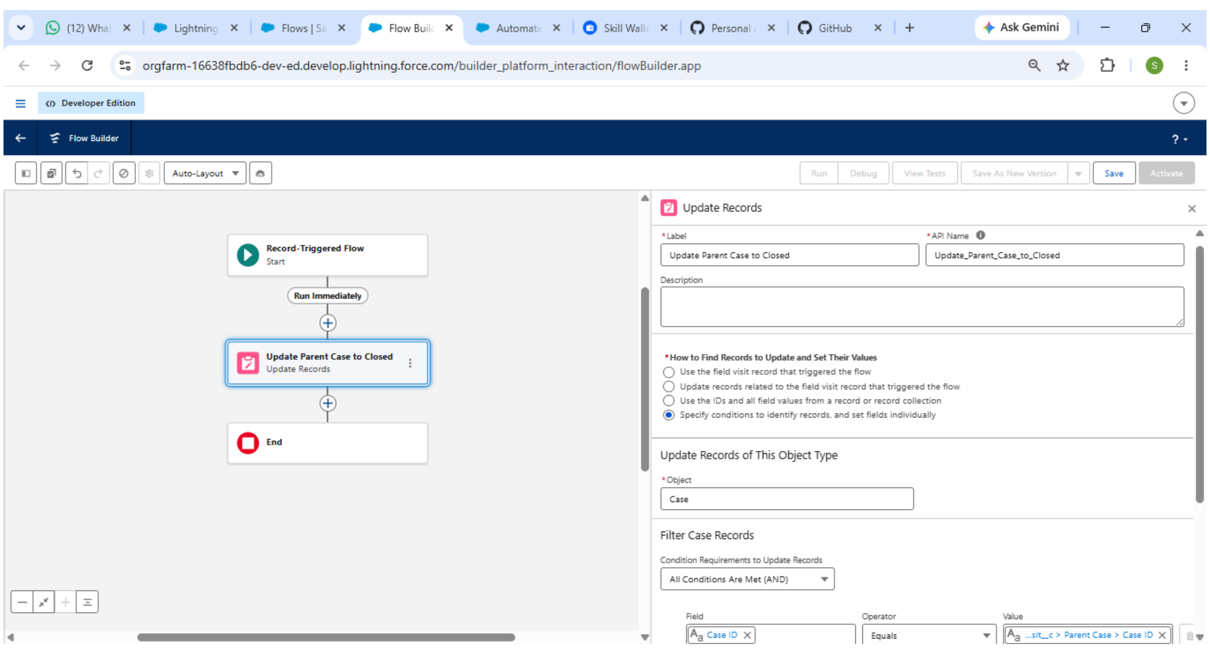
   - **Object:** Case

5. Configure the filter:

   - **Field:** Id
   - **Operator:** Equals
   - **Value:** {!$Record.Parent_Case__r.Id}

6. Configure field updates:

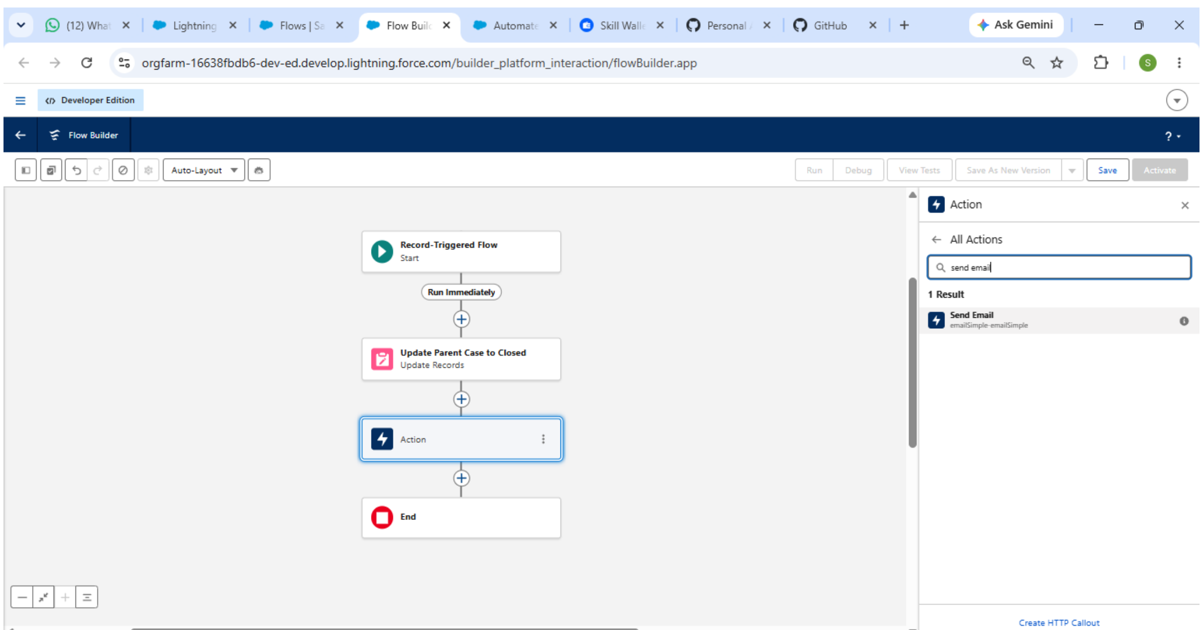
   - **Status** ← Closed
   - **Description** ← {!$Record.Technician_Notes__c}



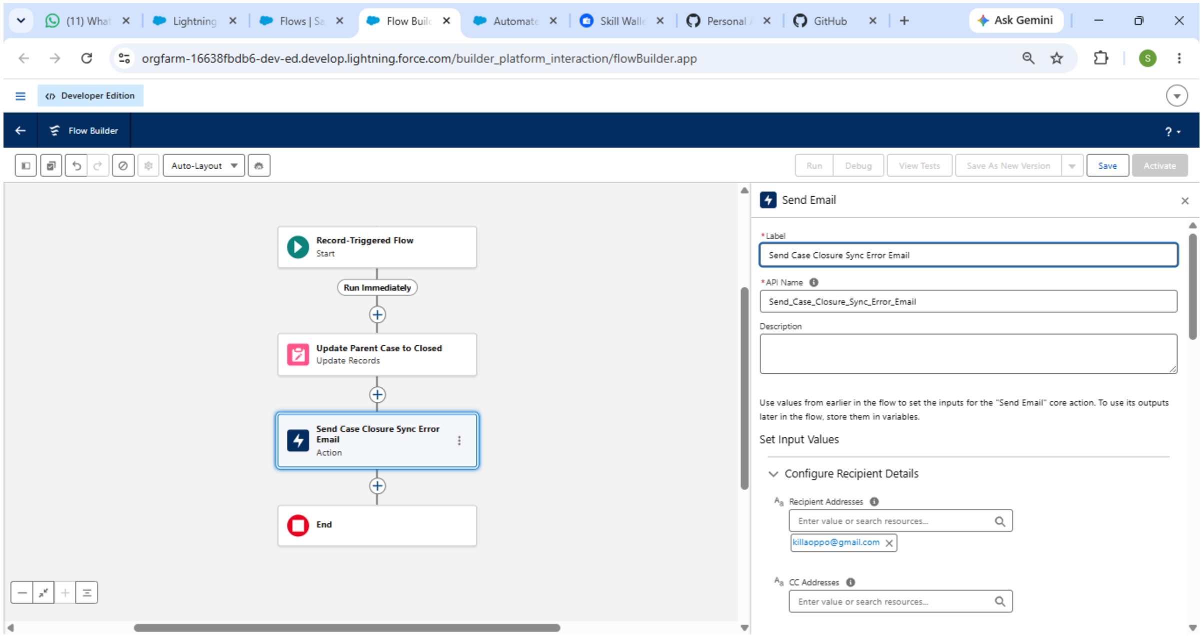
7. Save the configuration.

Step 8: Configure Fault Path Error Handling for Flow 2

1. Select **Update Parent Case to Closed**.
2. Choose **Add Fault Path**.
3. Click the plus (+) icon on the fault branch.
4. Select **Action**.
5. Search for and select **Send Email**.



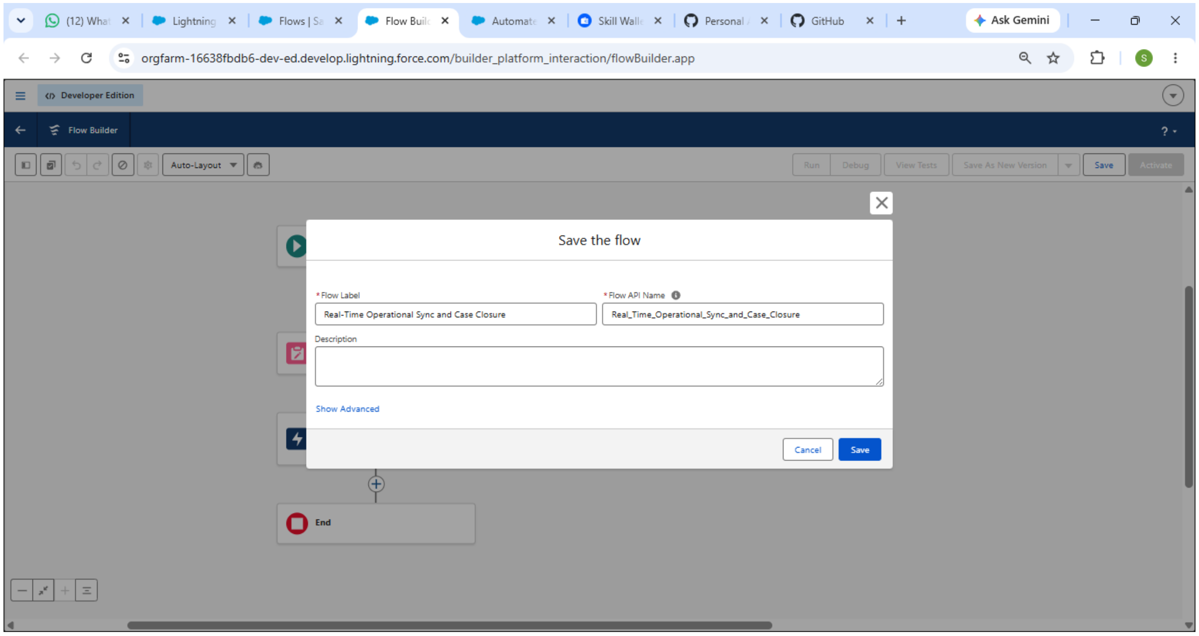
6. Configure:
   - **Label:** Send Case Closure Sync Error Email
   - **API Name:** Send_Case_Closure_Sync_Error_Email
   - **Body:** The Operational Sync Flow failed to close the parent Case record. Error Details: {!$Flow.FaultMessage}
   - **Subject:** CRITICAL ALERTER: GreenGrid Utilities Case Closure Failure
   - **Email Addresses:** Enter the administrator email address.



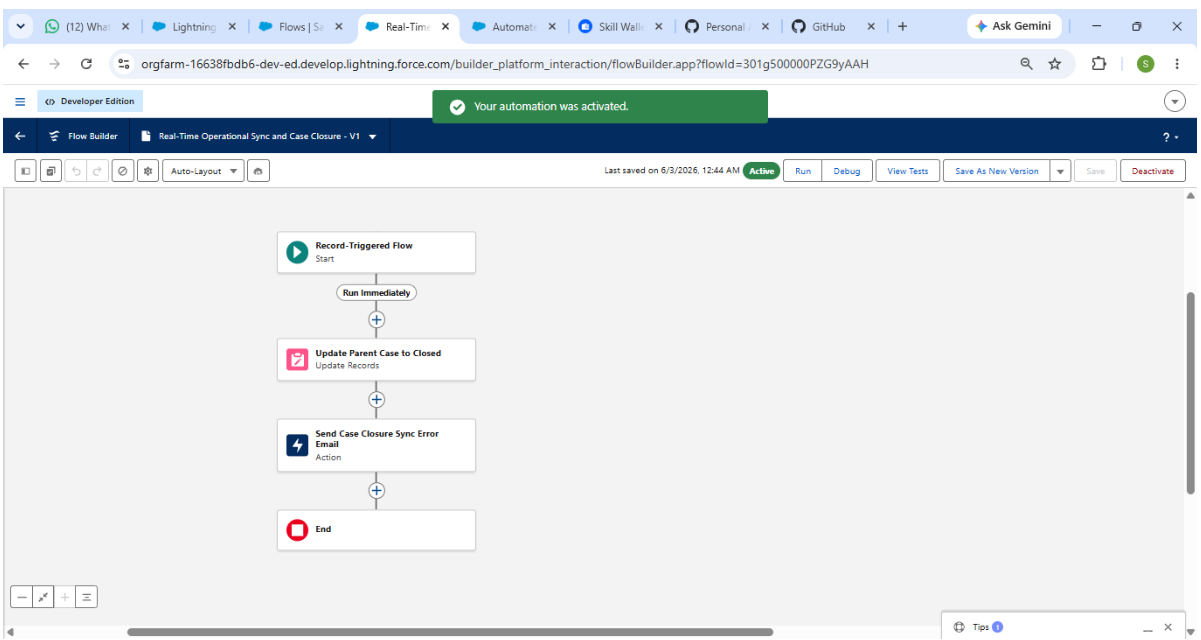
7. Click **Done**.

Step 9: Save and Activate Flow 2

1. Click **Save**.
2. Set the **Flow Label** to *Field Visit: Real-Time Operational Sync and Case Closure*.
3. Verify the **Flow API Name** is *Field_Visit_Real_Time_Operational_Sync_and_Case_Closure*.



4. Click **Save**.



5. Select **Activate**.

Step 10: Perform End-to-End Functional Testing

1. Open the **Cases** tab from the App Launcher. Create a test Case with Priority = High and Status = New, associate it with an existing Account and Utility_Asset__c record, and save it.

2. Confirm that a related **Field Visit** record is automatically generated under the Case's related list, indicating that **Flow 1** executed successfully.

3. Open the generated **Field Visit** record, click Edit, enter information in **Technician_Notes__c** (for example, *"Grid line 4 repair completed safely."*), change **Repair_Status__c** to **Resolved**, and save the record.

4. Return to the parent **Case** record and verify that the status has automatically changed to **Closed** and that the technician notes appear in the Case details, confirming successful execution of **Flow 2**.

Conclusion

Following the successful implementation of **Milestone 6: Automation Design**, GreenGrid Utilities has completed its transition from manual, paper-based operational processes to a responsive and fully automated digital environment.

Through the use of record-triggered automation, structured data validation, and comprehensive error-handling mechanisms, the solution fulfills operational efficiency requirements. The automated framework maintains data accuracy, reduces field service delays, safeguards critical records, and delivers real-time operational insights to management across the organization.